

Problem 10.1 (MATLAB)

Two-population classification using LDA and QDA

The file '**dataset_problem_10_1.mat**' contains size measurements for 37 male (coded with "1") and 29 female (coded with "2") individuals of a particular species of water snakes. The bivariate size data available in columns 2 and 3 in the data matrix, X , is: x_1 = tail length [mm], x_2 = snout to vent length [mm].

Since definitive determination of the gender of water snakes is a somewhat complicated procedure, it is desired to investigate whether a simple discrimination might be possible, based solely on tail and snout to length measurement of a particular snake with unknown gender.

Therefore: in this problem consider the 37 snakes known to be male as members of population 1 and likewise the 29 snakes known to be female as members of population 2. In the following, classification rules based on a Linear Discriminant Analysis (LDA) as well as a Quadratic Discriminant Analysis (QDA) will be constructed and performance evaluated assuming equal priors for males and females and equal misclassification "costs"

Data preparation and plot of observations

- Extract the size data for the male and female snakes from the data matrix
- Make a scatter plot of the bivariate observations from both populations in the same figure using different colors for males and females
- Calculate descriptive statistics for both populations, i.e. sample mean vectors and sample covariance matrices

Multivariate (bivariate) normal model check

- For both populations perform a check of the MVN assumption that will be used in the following, that is, make scatter matrix plots and QQ-plots for sample Mahalanobis distances as in earlier problems
- Conclude (hopefully) that the MVN assumptions are reasonably verified considering the relative small datasets

LDA or QDA most appropriate ?

- Perform the usual Bartlett test of a hypothesis of equal covariance matrices for the two populations
 - The low p-value for this test should indicate that the covariance matrices most probably are not equal, and thus QDA should be the most appropriate classification approach. However for the sake of comparison between LDA and QDA, in this problem, first LDA will be studied followed by QDA.
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LDA case

- First, test if the two populations are significantly separated in terms of mean vectors, that is, make a Hotelling T^2 test of a hypothesis of equal mean vectors. Remember that in the LDA case, a pooled estimate of the common covariance matrix for the two populations should be used, since their covariance matrices are assumed – somewhat against the above result - to be equal.
- Conclude that the populations indeed are significantly separated (that is, rejecting the hypothesis), thus confirming the visual impression from the first figure. Therefore, classification and discriminant analysis based on the two measured variables seems relevant

For the LDA case, the usual classification rule for equal priors and misclassification costs and observation $\mathbf{x}$

$$(\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2)^T \mathbf{S}_p^{-1} \left[\mathbf{x} - \frac{1}{2}(\bar{\mathbf{x}}_1 + \bar{\mathbf{x}}_2) \right] \geq \frac{1}{2} 0$$

can be reformulated using linear discriminant functions

$$d_i^{\text{LDA}}(\mathbf{x}) = \bar{\mathbf{x}}_i^T \mathbf{S}_p^{-1} \mathbf{x} - \frac{1}{2} \bar{\mathbf{x}}_i^T \mathbf{S}_p^{-1} \bar{\mathbf{x}}_i, \quad i = 1, 2$$

simply as

$$d_1^{\text{LDA}}(\mathbf{x}) \geq d_2^{\text{LDA}}(\mathbf{x})$$

- Calculate the discriminant functions $d_1^{\text{LDA}}(\mathbf{x})$, $d_2^{\text{LDA}}(\mathbf{x})$ as linear functions of the variables $x_1 = \text{tail length [mm]}$, $x_2 = \text{snout to vent length [mm]}$.
- Make a new scatter plot of the bivariate observations from both populations in the same figure and indicate the classification border (line) given by $d_1^{\text{LDA}}(\mathbf{x}) = d_2^{\text{LDA}}(\mathbf{x})$

Performance evaluation of LDA classification

- Calculate the confusion matrix for the known training data from the two populations using the constructed LDA classification and verify the number of misclassifications with the figure just made
- Calculate the Apparent Error Rate, APER, using the training data
- Calculate the confusion matrix and the estimated actual error rate, AER, for training data using the cross-validation (hold-out) procedure
- Compare APER and AER

Classification of new observations using LDA

Consider a new observation, $\mathbf{x}_{01} = \begin{bmatrix} 170 \\ 550 \end{bmatrix}$, for a snake with unknown gender

- Calculate the discriminant functions $d_1^{\text{LDA}}(\mathbf{x}_{01})$, $d_2^{\text{LDA}}(\mathbf{x}_{01})$ and classify the new observation as being male or female
- Calculate the posterior probability for the new observation to be male or female, respectively, which in the case of equal priors can be calculated as

$$P(1|\mathbf{x}_{01}) = \frac{\hat{f}_1(\mathbf{x}_{01})}{\hat{f}_1(\mathbf{x}_{01}) + \hat{f}_2(\mathbf{x}_{01})}$$

$$P(2|\mathbf{x}_{01}) = \frac{\hat{f}_2(\mathbf{x}_{01})}{\hat{f}_1(\mathbf{x}_{01}) + \hat{f}_2(\mathbf{x}_{01})}$$

where $\hat{f}_1(\mathbf{x}_{01})$ and $\hat{f}_2(\mathbf{x}_{01})$ are the “likelihoods”, that is the value of the MVN pdf’s for $\mathbf{x} = \mathbf{x}_{01}$ using sample mean vectors and common pooled sample covariance matrix

- Classify the new observation again, now using the MAP (Maximum Aposteriori) criterion, compare with above.
- Do the same for new observations, $\mathbf{x}_{02} = \begin{bmatrix} 200 \\ 600 \end{bmatrix}$ and $\mathbf{x}_{03} = \begin{bmatrix} 120 \\ 380 \end{bmatrix}$

QDA case

Now, turning to the case that the previously performed Bartlett test indicated as being most appropriate, namely the case where the covariance matrices for the two populations are not assumed to be equal – that is, the quadratic discriminant analysis case (QDA)

- First, make a renewed test if the two populations are significantly separated in terms of mean vectors, that is, make a Hotelling T^2 test of a hypothesis of equal mean vectors. Remember that in the QDA case, a pooled estimate of the common covariance matrix for the two populations can not be used, since their covariance matrices are assumed to be different, therefore only an approximate (large sample) test statistic is available.
- Conclude that the populations indeed are significantly separated (that is, again rejecting the hypothesis of equal mean vectors).

For the QDA case, the usual classification rule for equal priors and misclassification costs and observation $\mathbf{x}$

$$-\frac{1}{2}\mathbf{x}^T(\mathbf{S}_1^{-1} - \mathbf{S}_2^{-1})\mathbf{x} + (\bar{\mathbf{x}}_1^T\mathbf{S}_1^{-1} - \bar{\mathbf{x}}_2^T\mathbf{S}_2^{-1})\mathbf{x} - K \geq \frac{1}{2} 0$$

where

$$K = \frac{1}{2}\log\frac{|\mathbf{S}_1|}{|\mathbf{S}_2|} + \frac{1}{2}(\bar{\mathbf{x}}_1^T\mathbf{S}_1^{-1}\bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2^T\mathbf{S}_2^{-1}\bar{\mathbf{x}}_2)$$

can be reformulated using quadratic discriminant functions

$$d_i^{\text{QDA}}(\mathbf{x}) = -\frac{1}{2}\mathbf{x}^T\mathbf{S}_i^{-1}\mathbf{x} + \bar{\mathbf{x}}_i^T\mathbf{S}_i^{-1}\mathbf{x} - \frac{1}{2}\log|\mathbf{S}_i| - \frac{1}{2}\bar{\mathbf{x}}_i^T\mathbf{S}_i^{-1}\bar{\mathbf{x}}_i, \quad i = 1, 2$$

simply as

$$d_1^{\text{QDA}}(\mathbf{x}) \geq d_2^{\text{QDA}}(\mathbf{x})$$

- Calculate the discriminant functions $d_1^{\text{QDA}}(\mathbf{x})$, $d_2^{\text{QDA}}(\mathbf{x})$ as quadratic functions of the variables $x_1 = \text{tail length [mm]}$, $x_2 = \text{snout to vent length [mm]}$.
- Make a new scatter plot of the bivariate observations from both populations in the same figure and indicate the QDA classification border given by $d_1^{\text{QDA}}(\mathbf{x}) = d_2^{\text{QDA}}(\mathbf{x})$

Performance evaluation of QDA classification and comparison with LDA

- Calculate the confusion matrix for the known training data from the two populations using the constructed QDA classification and verify the number of misclassifications with the figure just made
- Calculate the Apparent Error Rate, APER, for QDA using the training data
- Calculate the confusion matrix and the estimated actual error rate, AER, for training data using the cross-validation (hold-out) procedure
- Compare APER and AER for QDA
- Compare APER and AER using the LDA and QDA classification, respectively, and comment

Classification of the new observations using QDA comparing with LDA

Consider the same new observation as for LDA, $\mathbf{x}_{01} = \begin{bmatrix} 170 \\ 550 \end{bmatrix}$, for a snake with unknown gender

- Calculate the discriminant functions $d_1^{\text{QDA}}(\mathbf{x}_{01})$, $d_2^{\text{QDA}}(\mathbf{x}_{01})$ and classify the new observation as being male or female
- Calculate the posterior probability for the new observation to be male or female, respectively, which in the case of equal priors can be calculated as

$$P(1|\mathbf{x}_{01}) = \frac{\hat{f}_1(\mathbf{x}_{01})}{\hat{f}_1(\mathbf{x}_{01}) + \hat{f}_2(\mathbf{x}_{01})}$$
$$P(2|\mathbf{x}_{01}) = \frac{\hat{f}_2(\mathbf{x}_{01})}{\hat{f}_1(\mathbf{x}_{01}) + \hat{f}_2(\mathbf{x}_{01})}$$

where $\hat{f}_1(\mathbf{x}_{01})$ and $\hat{f}_2(\mathbf{x}_{01})$ are the “likelihoods”, that is the value of the MVN pdf’s for $\mathbf{x} = \mathbf{x}_{01}$ using sample mean vectors and sample covariance matrices

- Classify the new observation again, now using the MAP (Maximum A posteriori) criterion, compare with above.
 - Compare the classification using QDA with the previously classification using LDA and comment
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- Do the same for the two other new observations, $\mathbf{x}_{02} = \begin{bmatrix} 200 \\ 600 \end{bmatrix}$ and $\mathbf{x}_{03} = \begin{bmatrix} 120 \\ 380 \end{bmatrix}$
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